

Nilay Toshniwal

Electronics & Communication Engineering, BITS Pilani Goa · Class of 2027

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EDUCATION

Birla Institute of Technology & Science, Pilani · K. K. Birla Goa Campus

2023 – 2027

B.E. (Hons.) Electronics & Communication Engineering, minor in Finance · CGPA 7.18 / 10

CERTIFICATIONS

Anthropic Academy AI Fluency: Framework & Foundations · AI Fluency for Builders · Claude Code in Action · Teaching AI Fluency

SKILLS

Agents and LLMs Claude Code (about 30 custom skills, MCP servers and scheduled agents of my own), Claude API, tool-calling agents, evaluation harnesses and failure taxonomies, LangChain, FAISS, Hugging Face

Machine learning PyTorch, scikit-learn, OpenCV, YOLO, Sentence-BERT, model quantisation, NumPy, SciPy, pandas, MLflow

Software Python, C++, C, SQL, Bash, Docker, Git, Linux, React, Next.js, Streamlit

Community and operations Sponsorship and partnerships, social media growth, event budgeting, vendor and cost negotiation

CAMPUS LEADERSHIP

Core Member, then Finance Head, DoFAM, BITS Goa

Aug 2024 – May 2026

- Ran the money for three campus fests a year. Waves, Quark and Spree, roughly **₹4–5 crore** combined, about USD 500K, split department by department against a fixed ceiling.
- Forty to fifty people** on the core team, each department certain its claim was the urgent one. Most of the job was saying no with a reason attached.

Sponsorship Lead and Core Member, Society of Automotive Engineers, BITS Pilani

Aug 2024 – Aug 2025

- Raised **₹25L+** across two vehicle builds, about USD 28K. Nearly all of it began as cold outreach that had to become a case a sponsor could sign.
- Ran the team's social accounts and grew the Instagram past **1,000 followers**, up from around 600. That is where most of those sponsor conversations started, so the posting and the fundraising were one job.

EXPERIENCE

Research and Development Engineering Intern, ElasticRun

Jun 2026 – Aug 2026

- Built a bimanual robot platform in ROS2 and MuJoCo, then traced a violent shaking in the arm to my own physics settings fighting the simulator, not a tuning problem. Residual movement dropped from **7.27 to 0.076 rad/s**.
- Audited the company's production warehouse simulator against its spec. **Seven engine-level defects**, every fix validated against a **1,341-case** regression suite.

PROJECTS

WINNER **PCB defect inspection, TATA Vayu AI Studio Hackathon national final**

- Led four people. Six defect types on bare PCBs with no golden board to compare against, so a new design costs nothing to inspect. Shipped as a GPU inference service.
- Every board comes out as release, rework or scrap with a rupee figure attached, not a box drawn on an image. It still finds the defects on board designs it has never seen, which is the hard case (**0.717 mAP@0.5**).

SECOND **Database reconstruction, APPCAIR JAW 2026 Hackathon**

- 687 documents**, no database and no schema. Rebuilt the store first, 155 works, 28 clients and 39 people across **41 tables and 9,973 rows**, then answered all **333 questions** from it. Scored **100.000**, the maximum on the official set.
- A classifier routes each question to one of **23 query shapes** and extracts its parameters; anything unmatched compiles into a filter-and-reduce over the whole store. **No generative model runs anywhere in the answer path**, so every figure is computed in Python over exactly-parsed integers.

market-query-agent, a tool-using agent and the harness that grades it

- Answers analytical questions about Indian equity data by writing and running SQL on `claude-haiku-4-5`. **45 of 48** graded questions correct, at **\$0.016** each.
- All three misses are in the one tier I wrote to catch them, questions the data cannot answer. Method is committed before results, so numbers cannot be fitted to it afterwards.

Autonomous options-trading agent, live on real market data

- Sells defined-risk SPY option premium only when volatility is measurably expensive. On its first live run it declined to trade, implied **12.81** against realised **13.28**. Every decision, refusals included, is signed into a log that verifies with one command.
- I wrote down what would count as evidence before any backtest code existed, provable from git history. The edge held over **1,741 observations** and seven years of unseen data, and **eleven numbered safety checks** cut the worst historical loss from **4.38% to 1.39%**.
- One of my written-down predictions turned out wrong, and it sits in the repo with the ones that worked.

A processor I designed, and custom hardware to speed it up

- Measured before optimising. Running a small neural network took **139,813 clock cycles**, nearly all of it software multiplication because the chip had no hardware for it. I built that hardware and it came down to **4,241**, checked against a reference model.